TEMASEK JUNIOR COLLEGE

PRELIMINARY EXAMINATION

Higher 2

GEOGRAPHY

9173/02

Paper 2

11 September 2024

Additional Materials: Insert

3 hours

READ THESE INSTRUCTIONS FIRST

If you have been given an Answer Booklet, follow the instructions on the front cover of the Booklet.
Write your Centre number, index number and name on the work you hand in.
Write in dark blue or black pen on both sides of the paper.
You may use an HB pencil for any diagrams or graphs.
Do not use staples, paper clips, glue or correction fluid.

Answer Question 1 in **Section A**.
Answer Question 2 in **Section B**.
Answer **one** question in **Section C**.

At the end of the examination, fasten all your work securely together.
The number of marks is given in brackets [ ] at the end of each question or part question.

This document consists of **4** printed pages

Answer Questions 1 and 2 and
EITHER Question 3 **OR** Question 4

Section A

Cluster 4 Fieldwork

1 As global temperatures rise and extreme weather events become more frequent, the impact of climate change is increasingly felt in urban areas, particularly in the form of the Urban Heat Island (UHI) effect. A group of four students carried out an investigation to study the relationship between the UHI effect and willingness for climate change action among residents in different parts of Singapore. The hypothesis guiding this investigation is:

'Residents living in areas with higher Urban Heat Island Vulnerability Index are more willing to adopt measures to mitigate climate change.'

Two sites with differing levels of UHI vulnerability were selected for comparison, and a total of 50 questionnaires were administered — 25 at each site. The four students carried out the survey simultaneously on a weekday Thursday afternoon between 2 to 4pm in October by approaching every third person encountered at each site shown in Resource 2. This was followed by conducting in-depth interviews with ten respondents on a separate day to gather insights into their attitudes and willingness to take action against climate change.

Resource 1 shows the map of urban heat vulnerability index in Singapore. Resources 2A and 2B shows the sampling locations of Bukit Merah and Serangoon respectively. Resource 3 shows the survey questionnaire used in the investigation. Resource 4 shows selected data from the questionnaire survey.

(a) Explain why the hypothesis is appropriate for the investigation. [4]

(b) With reference to Resources 2A and 2B, explain how the students could minimise potential risks during the investigation. [6]

(c) Explain **three** criteria the students could use to select respondents for the interview. [5]

(d) With reference to Resource 3, explain how the design of the survey questionnaire used to collect data could be improved. [5]

(e) Using Resource 4, explain whether the results support the hypothesis. [5]

(f) Use an appropriate data representation technique to represent the data shown in Resource 4. [5]

(g) Evaluate the validity of the students' investigation. [10]

Section B

Cluster 3 Sustainable Future and Climate Change

2 Resource 5 shows the change in urban liveability scores. Resource 6 shows an aerial photograph of Vienna's suburbs. Resource 7 shows regional variations in gender disparity in completing school education in India. Resource 8 shows perceived seriousness towards climate change between men and women and GDP per person.

(a) Using Resource 5, describe the main patterns of change in urban liveability scores of cities in the world. [5]

(b) With reference to Resource 6, suggest why Vienna has high urban liveability. [4]

(c) Using Resource 7, describe the regional variations in gender disparity in completing school education in India. [5]

(d) Explain the importance of gender equality in progressing towards liveable cities. [4]

(e) Explain why strategies to manage the well-being of women living in cities are not always successful. [6]

(f) Suggest reasons to account for the differences in perceived seriousness towards climate change shown in Resource 8. [6]

Section C

Answer **EITHER** question 3 **OR** question 4.

Cluster 3 Sustainable Future and Climate Change

3 'The effects of contemporary climate change vary greatly'.

Evaluate this statement. [20]

4 'Sustainable urban development is difficult to achieve.'

Evaluate this statement. [20]

TEMASEK JUNIOR COLLEGE

Preliminary Examination

Higher 2

GEOGRAPHY

9173/02

Paper 2

11 September 2024

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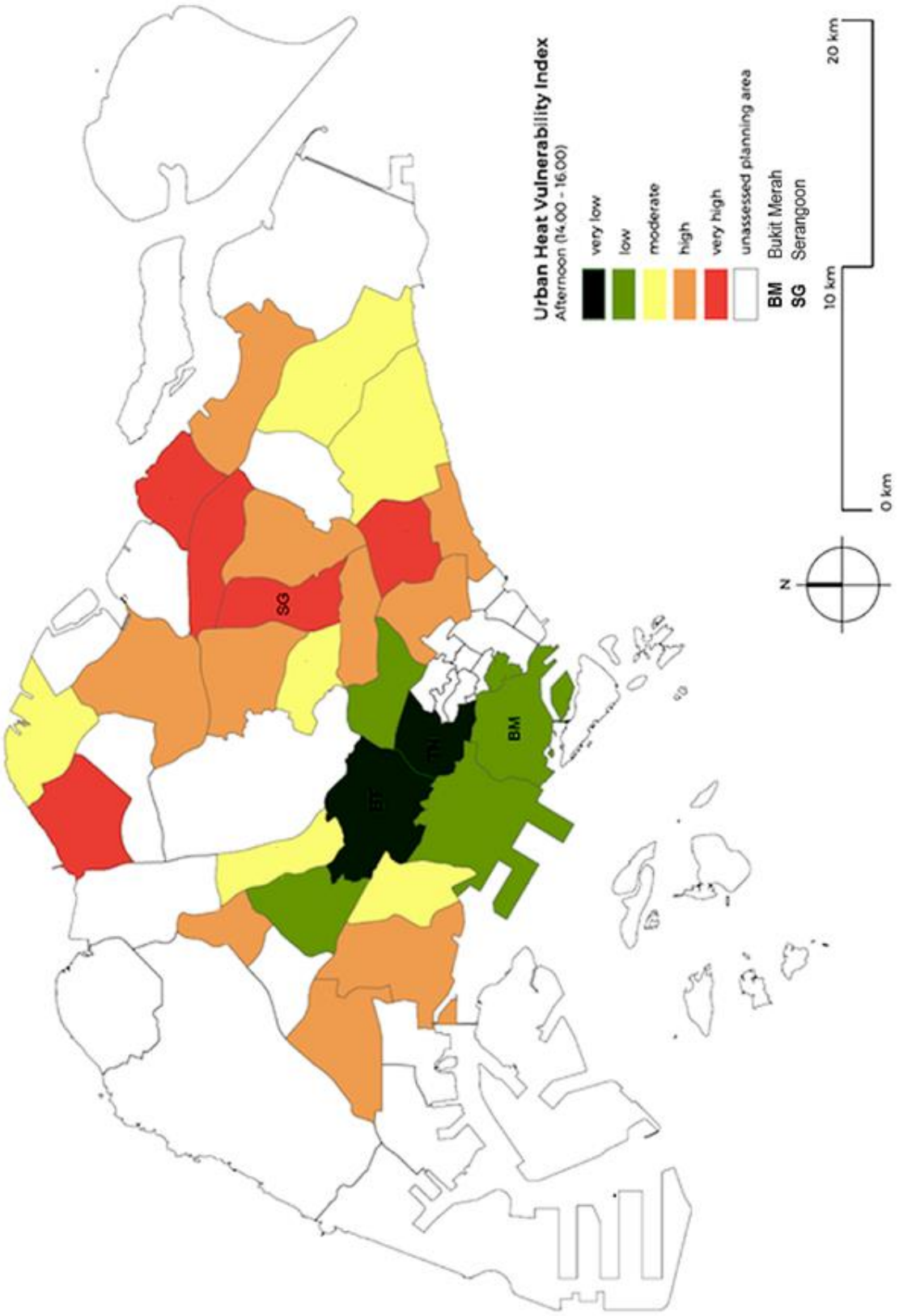
3 hours

READ THESE INSTRUCTIONS FIRST

This Insert contains all the Resources referred to in the questions.

This document consists of **9** printed pages and **3** blank pages.

Resource 1 for Question 1
Map of Urban Heat Vulnerability Index in Singapore



Resource 2A for Question 1

Sampling Location A - Bukit Merah



Resource 2B for Question 2

Sampling Location B – Serangoon



Resource 3 for Question 1

Survey Questionnaire

1. Are you a resident in this neighbourhood? Yes/ No

2. Have you, in the last 5 years, experienced any form of heat stress in your neighourhood? Yes/No

3. Do you feel that the pattern of weather is generally changing? Yes/No/Don't know

4. Have you heard of climate change? Yes/No/Don't know

5. Do you think climate change is something that is affecting or is going to affect you, personally? Yes/No/Don't know

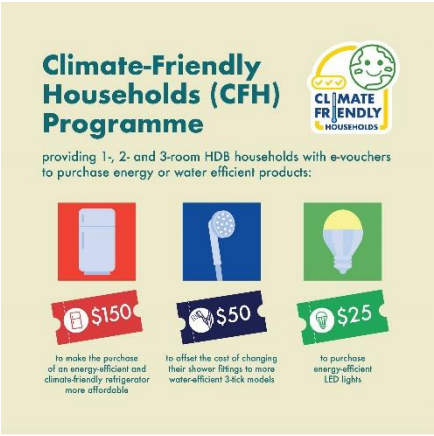
6. Have you ever taken, or do you regularly take, any action out of concern for climate change? Yes/No/Don't know

7. Please indicate how much you agree or disagree with the following statements on climate change

	1 Strongly Agree	2 Agree	3 Neutral	4 Disagree	5 Strongly disagree
We can all do our part to reduce the effects of climate change.					
It is already too late to do anything about climate change.					
Human activities have no significant impact on global temperatures.					
Claims that human activities are changing the climate are exaggerated.					

8. Have you heard of the *Climate Vouchers* given by the government to households (see picture below)?

(Yes/No/Don't know)



Resource 4 for Question 1

Selected tallied data from questionnaire

Bukit Merah

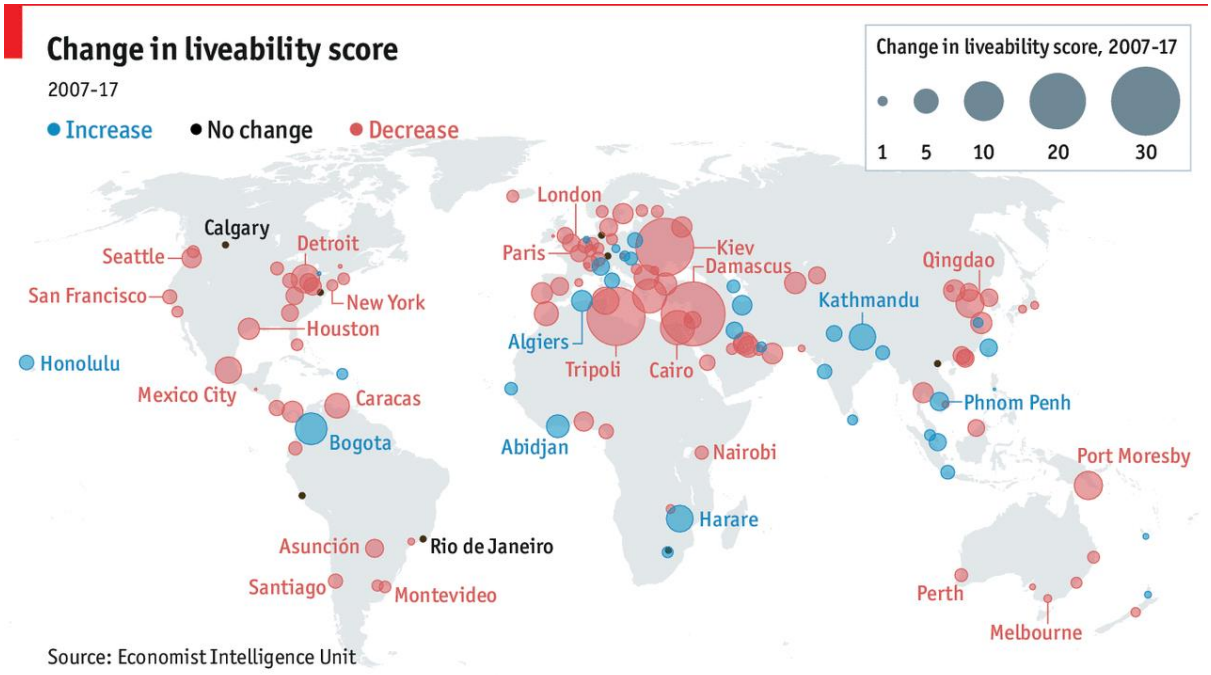
	1 Strongly Agree	2 Agree	3 Neutral	4 Disagree	5 Strongly disagree
We can all do every part to reduce the effects of climate change.	9	8	2	3	3
Human activities have no significant impact on global temperatures.	0	5	3	12	5

Serangoon Road

	1 Strongly Agree	2 Agree	3 Neutral	4 Disagree	5 Strongly disagree
We can all do our part to reduce the effects of climate change.	0	9	6	10	0
Human activities have no significant impact on global temperatures.	2	4	5	9	5

Resource 5 for Question 2

Change in urban liveability score



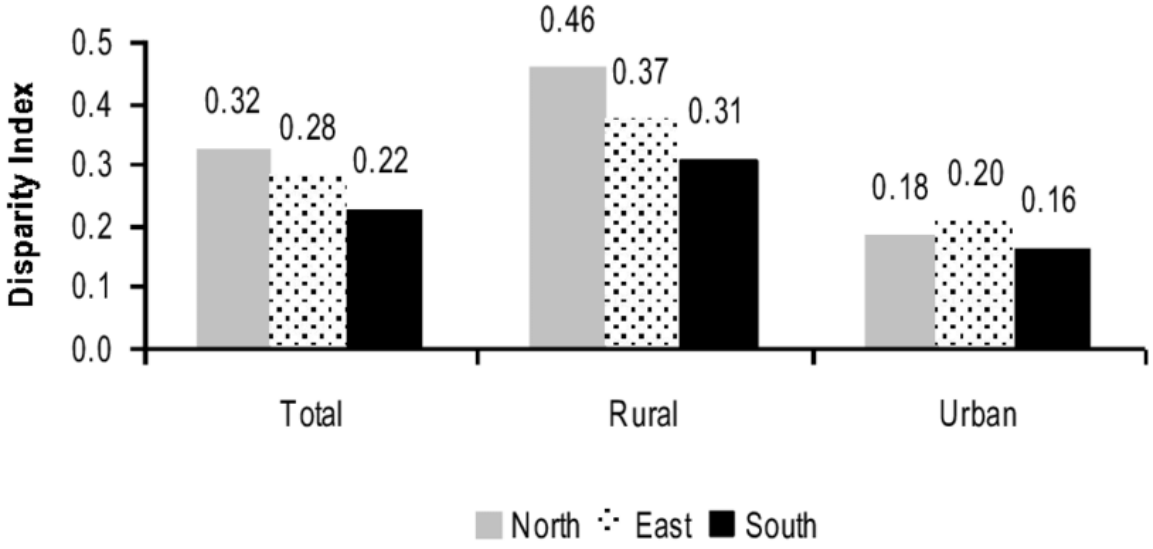
Resource 6 for Question 2

An aerial photograph of Vienna's suburbs



Resource 7 for Question 2

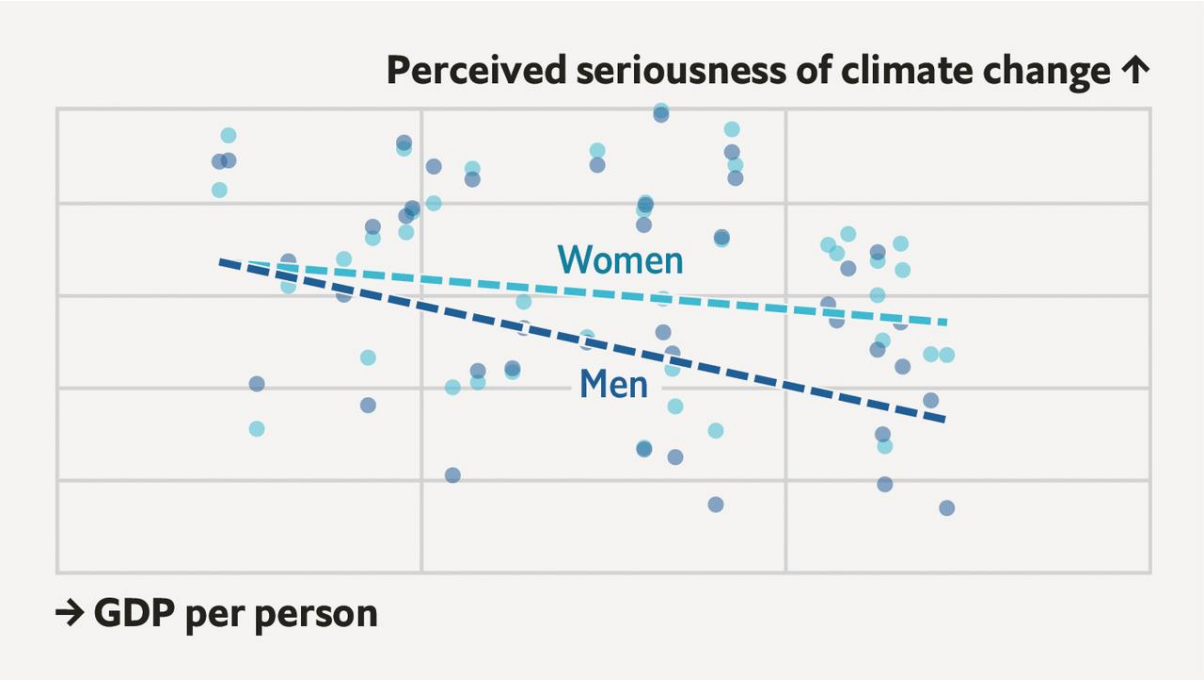
Regional variations in gender disparity in completing school education in India



The disparity index used here highlights women's disadvantages relative to men. Values closer to 0 indicate greater gender equality.

Resource 8 for Question 2

Perceived seriousness towards climate change between men and women and GDP per person



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